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Random forest as a potential multivariate method for near-infrared (NIR) spectroscopic analysis of complex mixture samples: Gasoline and naphtha

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ABSTRACT

Random forest (RF) has been demonstrated as a potential multivariate method for near-infrared (NIR) spectroscopic analysis of petroleum-driven products, highly complex mixtures of diverse hydrocarbons. For the study, a NIR dataset of gasoline samples and two separate NIR datasets of naphtha samples were prepared. These samples were carefully prepared over a long period to maximize compositional variation in each dataset. Partial least squares (PLS), the most widely adopted method in multivariate analysis, and RF were used to determine research octane numbers (RONs) of gasoline samples, and total paraffin, total naphthene and total aromatic concentrations of naphtha samples. The resulting accuracies of quantitative analysis for these samples were generally improved when RF was used. In addition, chance for overfitting of a model, which would occur occasionally in PLS modeling, was substantially lessened or possibly eliminated by the use of RF. On the contrary, in the case of RF, a calibration dataset composed of samples with narrow interval in property or concentration variation was required to improve the accuracy. Consequently, RF could be a useful multivariate method to analyze NIR as well as other spectroscopic data acquired from petroleum refining products, when properly utilized.

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1. Introduction

Near-infrared (NIR) spectroscopy is a widely adopted spectroscopic method for analysis of petroleum-driven products [1,2] because it is fast, non-destructive and capable of simultaneous multi-component analysis. Mostly, these samples are complex mixtures of enormously diverse hydrocarbons, so their NIR spectral feature is severely overlapped and broad without distinct signature of individual components. Under this situation, the use of multivariate calibration methods is necessary to establish correlation to describe the change of concentration (or property) from these highly overlapped spectra. Among them, partial least squares (PLS) regression has been the most routinely utilized in many practical analytical applications, while other chemometric methods have been also proposed [3–6].

Recently, random forest (RF), one of the latest ensemble methods in machine learning, has been recognized as an effective method for classification, especially in the fields of gene analysis, metabolomics and medical image analysis [7–11]. Analytical data acquired from microarrays, mass spectrometry and chromatography have been extensively employed for classification using RF [12–14]. RF is an ensemble classifier method that is an extension of decision tree concept based on classification and regression tree (CART) [15]. RF combines many trees to form a forest for analysis. An individual tree represents a model describing the characteristics of an input feature that is present in a subset of the whole dataset. Detail mathematical descriptions of RF can be found in other papers [15–18].

Although RF has been widely utilized for classification, it is also capable of quantitative analysis [19,20]. When a combined learner (forest) composed of multiple models (trees) is used instead of adopting a single model developed using a parametric method (such as PLS), the determination of concentrations would be more robust through the simultaneous estimation of concentrations from many independent trees. Each tree in a forest is developed by employing randomly selected samples in a given dataset, and variables (wavelengths or wavenumbers) used to build a tree are also randomly selected, so each tree can be regarded as independent. When many independent trees are combined for analysis, the risks of biased decisions or overfitting would greatly decrease. Nonetheless, a limited number of studies of RF based on NIR spectroscopy have been reported such

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as the following publication. RF was used to estimate soil organic carbon and clay content from NIR diffuse reflectance spectra of soils along with other chemometric methods [19]. Also, RF was employed to predict *Sirex noctilio*-induced water stress based on visible-NIR hyperspectral data acquired from *Pinus patios* [20]. As shown, the acquaintance of RF in the field of NIR spectroscopy is now in a beginning stage. The prospective merits of RF would be more effective when highly overlapping spectra obtained from complex samples are analyzed.

In this study, we have explored RF as a potential multivariate method for NIR analysis of petroleum-driven products, complex mixtures of hydrocarbons with similar molecular structures as well as isomers. For this purpose, gasoline and naphtha were chosen since these were one of most concern-centered samples requiring a fast and reliable quality control and assurance in refinery. Using PLS and RF, research octane number (RON) was determined for gasoline samples using the corresponding NIR spectra. Simultaneously, total paraffin, total naphthene and total aromatic concentrations of naphtha samples were determined using both methods. Then, the resulting accuracies from the use of PLS and RF were compared for both gasoline and naphtha samples. To understand the characteristics of RF modeling in the analysis of these samples, the variable importance indicating the significance of selected wavelengths to build individual trees was examined. In addition, the variation of accuracies, when the number of trees in a RF model as well as the number of training samples changed, were also investigated.

The detailed procedure of RF for quantitative analysis using spectroscopic data is described in the following section. Graphical representations, rather than direct mathematical descriptions, were used to make the RF modeling procedure more easily understandable for general analysts in the relevant fields.

2. Procedure for building a RF model using spectroscopic data

RF is a substantial modification of bagging that builds a large collection of de-correlated trees, and then averages them [15]. For regression, we simply fit the same regression tree many times to bootstrap-sampled versions of the training data, and average the results. It is to improve the variance reduction of bagging by reducing the correlation between the trees, without increasing the variance too much. That is, for $b = 1$ to B: (a) Draw a bootstrap sample Z^* of size N from the training data. (b) Grow a RF tree T_b to the bootstrapped data, by recursively repeating the following steps for each terminal node of the tree until the minimum node size n_{min} is reached, then produces the ensemble of trees $\{T_b\}$ ($b = 1$ to B). We can make a prediction at a new point x as an average of $\{T_b(x)\}$ for $b = 1$ to B [15].

Fig. 1 shows the graphical presentation detailing overall procedure for constructing a RF model for quantitative analysis using spectroscopic data. Suppose that a spectral dataset composed of n samples (spectra) with p wavelengths (variables) is used for analysis. For example, a dataset shown in the first row is composed of 12 spectra with 300 wavelengths. Each spectrum is shown in a different color for easier visual recognition. To build a tree, initially two thirds (66.7%) of n total spectra are randomly selected (8 spectra out of a total of 12 spectra in this scheme). This

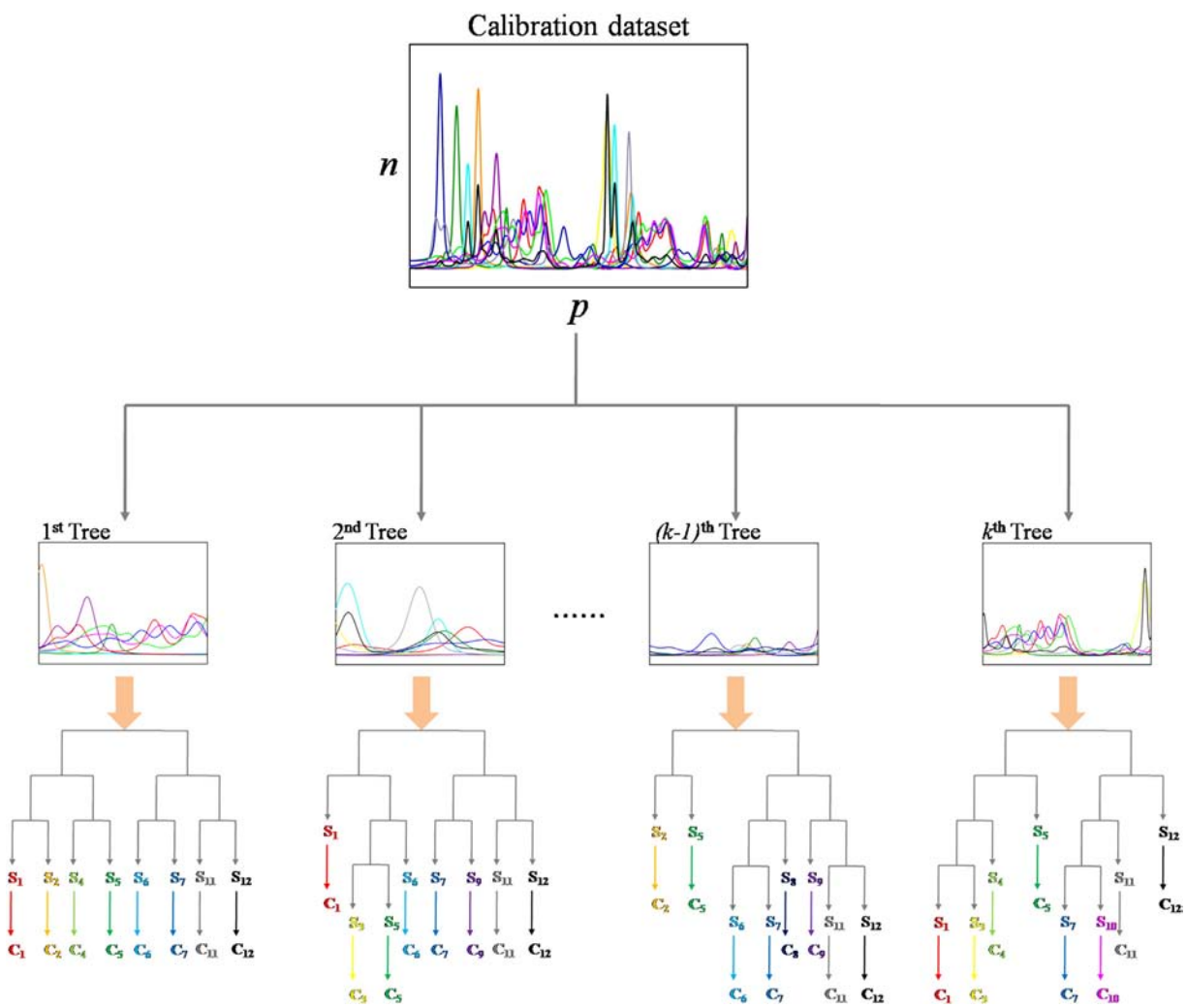

Fig. 1. Graphical presentation detailing the construction of a RF model for quantitative analysis using spectroscopic data.

procedure is called out-of-bag (OOB) sampling [15], and is similar to dividing a dataset for cross validation. The percentage of the sample selection could be changed, but the selection of two thirds out of the total samples is typical. After the random selection of samples, one third (33.3%) of the wavelengths out of p total wavelengths are randomly selected again (100 wavelengths out of a total of 300 wavelengths in this scheme) and are used to build a tree. Randomly selected wavelengths are graphically represented in the second row of Fig. 1.

When variables (absorbance values in the case of absorption-based spectroscopic data) are divided into two daughter nodes, a wavelength that can minimize the variances of divided daughter nodes is identified out of a randomly selected set of 100 wavelengths. The minimal sum of both variances corresponds to the more distinct two daughter nodes. Next two daughter nodes divided at the previous node are further classified by searching again for the best wavelengths out of the selected 100 wavelengths. The same procedure of building nodes is repeated until the selected samples for a given tree are fully classified. Finally, each classified sample is assigned to the corresponding reference concentration. For example, sample S_1 is matched to the corresponding concentration C_1. After building the first tree, the OOB samples that remain (one third of n spectra, 4 spectra in this scheme) are passed down the tree and their concentrations are determined. Then, the so called OOB error, $E_{X,Y}(Y - h(X))^2$, is obtained where $h(X)$ is a numerical predictor, X is a set of input variables and Y is an actual concentration [15].

Next, the second decision tree is built by employing the same procedure used to construct the first tree. Again, samples corresponding to two thirds of n total samples were randomly selected and were used to build the second tree. Since two trees have now been constructed, two predicted concentrations determined by two separate trees are available for each sample. The OOB error is again calculated using the average concentration of the two predicted concentrations. By repeating each tree building step, k trees are built and finally form a RF model, as shown in Fig. 1. Overall, the selected samples as well as the wavelengths employed in each tree will not be the same due to random selection. Therefore, each individual tree in a forest is regarded as an independent concentration predictor.

Fig. 2 shows the prediction procedure for a sample in an independent dataset, which is basically the same as the procedure used to predict OOB samples described above. For an unknown sample spectrum, the wavelengths employed in a given tree are again selected, and the corresponding absorbance values are passed through the tree to decide the concentration of a sample. The bold lines in each tree indicate the procedure of determining the concentration of a sample. Therefore, k concentrations individually determined by each tree are available for each sample. Ultimately, the final predicted concentration of an unknown sample ($C_{predicted}$) is obtained by averaging these k predicted values.

3. Experimental

All NIR spectra of gasoline samples (total of 379 samples) collected over nearly two years were acquired from a petroleum company in Korea. The main purpose of collecting these samples over two years was to prepare the dataset to include diverse origins of crude oils as

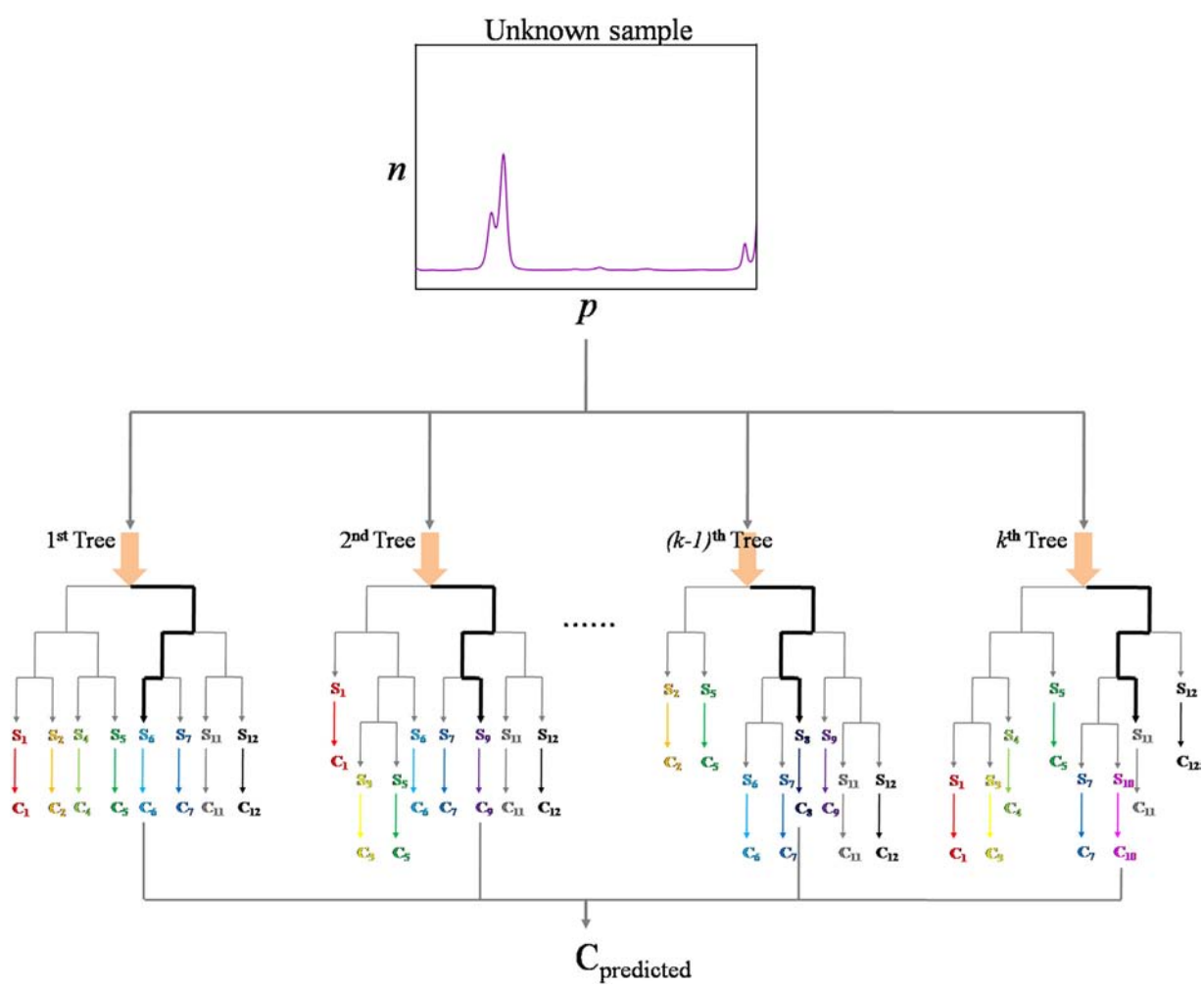

Fig. 2. Graphical presentation detailing the prediction procedure based on a developed RF model.

Table 1
The variation of RONs for gasoline samples, and total paraffin, naphthene and total aromatic concentrations for naphtha samples.

Dataset	Item	Maximum	Minimum	Average
Gasoline	RON	92.40	88.40	90.10
A	Total-paraffin (wt%)	87.16	75.65	82.17
	Total-naphthene (wt%)	15.61	9.49	12.35
	Total-aromatic (wt%)	7.76	2.95	5.03
B	Total-paraffin (wt%)	81.67	72.33	76.64
	Total-naphthene (wt%)	20.16	12.88	16.53
	Total-aromatic (wt%)	8.27	4.34	6.76

well as compositional variations based on the four seasonal grade changes. RON values of these samples were determined using a conventional knock engine [21]. Table 1 shows the variation of RON values in the dataset.

Two NIR datasets of naphtha samples, provided by two different petrochemical companies in Korea, were used. These are referred to as datasets A and B in this study. The samples in the datasets were cautiously collected over a one year period to maximize compositional variation by including diverse naphtha samples from different domestic and international suppliers. Table 1 also shows the variations in total paraffin, total naphthene and total aromatic concentrations present in both datasets. These concentrations were determined using a Hewlett-Packard 5890 GC equipped with three separation columns and a flame ionization detector (FID). The ranges of compositional variation in datasets A and B were slightly different, as shown in the table.

All three companies owned an FT-NIR spectrometer made by the same manufacturer (ABB Bomem MB-160, Quebec City, Canada). The spectrometers were equipped with tungsten–halogen sources and DTGS detectors. A 0.5 mm pathlength cell made of CaF_2 was used to collect sample spectra. Air was used as a background for all of the samples. NIR spectra were collected over 5000–4000 cm^{-1} at a resolution of 8 cm^{-1}.

All spectral processing, including baseline correction, normalization, PLS and RF was performed using MATLAB Version 7.0 (MathWorks Inc., MA, USA). For the gasoline dataset, 281 and 98 spectra were randomly divided into calibration and validation sets, respectively. In the case of naphtha samples, out of 201 total spectra, 160 and 41 spectra were assigned as calibration and validation sets, respectively, for dataset A. For dataset B, out of a total of 296 spectra, 238 and 58 spectra were assigned as calibration and validation sets, respectively.

4. Results and discussion

4.1. NIR spectral features of gasoline and naphtha samples

Fig. 3(a) shows normalized NIR spectra of all gasoline samples in the 4800–4000 cm^{-1} range. For normalization, the baselines of raw NIR spectra were linearly offset to zero at 4800 cm^{-1}, and the peak areas under the 4800–4000 cm^{-1} range were calculated. Each baseline-corrected spectrum was divided by the corresponding peak area. The 4700–4550 cm^{-1} range, where only NIR combination bands corresponding to aromatic hydrocarbons are present, is magnified. The combination bands of various hydrocarbons such as paraffin, naphthene and aromatic components appear in the 4500–4000 cm^{-1} range. Fig. 3(b) shows normalized NIR spectra of naphtha samples in the 4800–4000 cm^{-1} range. Red and blue colors indicate NIR spectra in datasets A and B, respectively. The 4700–4550 cm^{-1} range is also highlighted.

The spectral features of gasoline samples are slightly different from those of naphtha samples. Gasoline contains more aromatic hydrocarbons to boost RON value, so obviously the corresponding absorption of aromatic components in the 4700–4550 cm^{-1} range is greater compared to that of naphtha. By contrast, the bands of gasoline are relatively broader compared to those of naphtha, indicating severe spectral

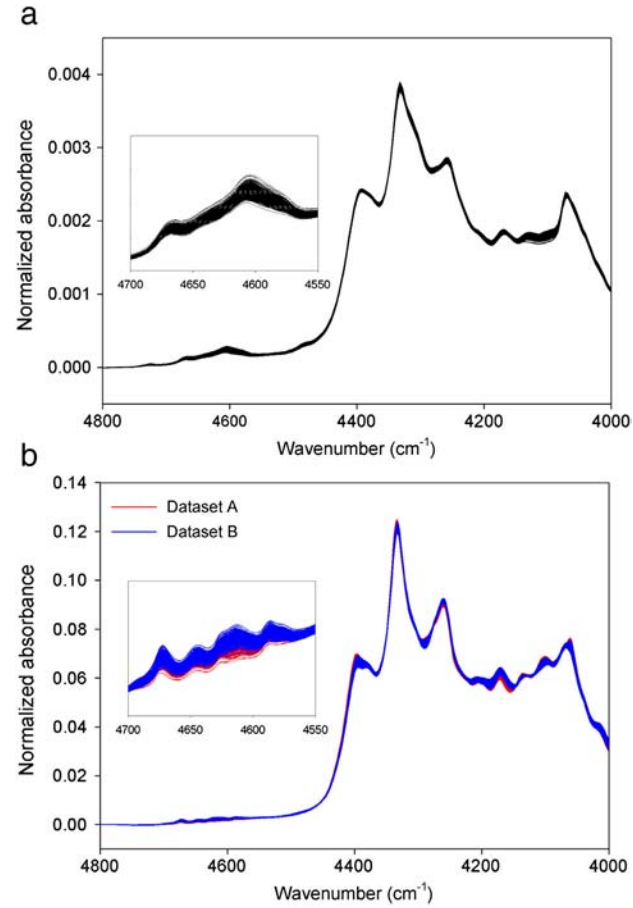

Fig. 3. Normalized NIR spectra of all gasoline (a) and naphtha samples (b) in the 4800–4000 cm^{-1} range. The 4700–4550 cm^{-1} range is highlighted for both samples. Red and blue colors indicate NIR spectra in naphtha datasets A and B, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

overlapping due to the higher degree of compositional complexity in gasoline. As generally known, gasoline is a blended product of diverse feed stocks such as reformate, heavy aromatic components, light straight run (LSR) and so on; while, naphtha is a group of hydrocarbons (composed of ~5–10 carbon atoms) from fractional distillation of crude oils at temperature ranging approximately from 50 °C to 160 °C. In the case of naphtha, NIR spectral features from both datasets are quite similar, since the compositions of both naphtha samples are not significantly different, as shown in Table 1. Since the NIR spectral features of both gasoline and naphtha samples are the results of extensive spectral overlapping of individual components, multivariate methods are necessary to determine their physical property or chemical composition.

4.2. Comparison of PLS and RF for the determination of naphtha composition

PLS was initially used to determine RONs for gasoline samples, and total paraffin, total naphthene and total aromatic concentrations for naphtha samples. In both cases, full spectral range (4800–4000 cm^{-1}) was used for PLS. For cross validation, 75% and 25% of samples in the calibration set were further divided into new calibration and validation sets, respectively. The random assignment of samples into calibration and validation sets was repeated 200 times for cross validation. Using the developed PLS models by cross validation, RONs or concentrations of samples in the corresponding original validation sets were predicted, and the resulting standard error of predictions (SEPs) were acquired.

In the case of gasoline samples, the resulting SEC (standard error of calibration) and SEP were 0.19 and 0.27, respectively. Five factors were used for the PLS modeling. The SEP of 0.27 is comparable with the results from other publications [22]. Results for the determination of total paraffin, total naphthene and aromatic concentration using datasets A and B using PLS are summarized in Table 2. The SEPs resulting from dataset A are relatively worse than those from dataset B. It would be difficult to address the exact causes; however, it is most probably due to differences in the accuracy of the reference methods used for the two datasets. Although the same reference method based on gas chromatography was used for both datasets, the reproducibility of measurement may depend on several factors such as assignment of chromatographic peaks, separation efficiency, instrumental conditions and the level of operator skill. Since naphtha composition is also highly complex with more than 100 individual components, many chromatographic peaks are located close to each other. Therefore, even slight variations in retention time could influence the accuracy of the compositional analysis. It is reasonable to hypothesize that the GC method used to determine reference concentrations in dataset A is less accurate than that used for dataset B.

Determination of an optimal number of factors in PLS is required, since a model could be possibly overfit when excess factors are used, while, a RF model would not be overfit even with the inclusion of large number of trees. This non-overfitting nature of RF has been mathematically proven [15]. Fig. 4 shows the variation in SEP as a function of the number of trees (up to 2000 trees) in RF modeling for the determination of RONs (a) for gasoline samples, and total paraffin (b), total naphthene (c) and total aromatic (d) concentrations for naphtha samples. For detail examination, the range of tree number from 200 to 2000 is highlighted in each plot.

In the case of determining RON using RF, the resulting SEPs are relatively high when the number of trees used in an RF model is below 200, indicating poor prediction ability. By continuously adding more trees, the SEPs oscillate marginally such as until 1700 trees and then remain nearly constant up to 2000 trees. The addition of trees to a model can be viewed as the selection of more independent judges (trees) to make decisions about a case. As long as the added judges are independent and selected by random fashion without any bias, the final decision that reflects all votes from each judge will not be biased. As explained in Fig. 1, individual trees in a forest are independent. For the determination of RONs, 2000 trees were incorporated into the RF model since the SEPs sufficiently converge to a constant level.

In the determination of naphtha compositions using RF, the SEP variation patterns as the increase of tree numbers are generally similar to those for the determination of RON. In all three cases, the SEPs sufficiently converged when 2000 trees were used. So, 2000 trees were incorporated into the RF models for the determination of total paraffin, total naphthene and total aromatic concentrations.

With the use of RF, the resulting SEC and SEP for the determination of RON were 0.17 and 0.22, respectively. The improvement in accuracy is significant compared with the SEP resulted using PLS, since it is larger than 0.3, the reproducibility of RON measurement by a knock engine (ASTM D2699). It indicates that RF is effective to describe the variation of RONs from the highly overlapped NIR spectra of gasoline samples as shown in Fig. 3(a).

In the case of using naphtha dataset A, the SECs derived using both methods are quite similar, while the improvement in SEPs is relatively more significant when RF is used. In addition, the SEPs are greater than the corresponding SECs in the case of PLS. This indicates that the PLS models could not fully account for the spectral variation present in the validation set, so the increase in the resulting SEPs is possibly due to overfitting. Meanwhile, in the case of RF, the differences between SEC and SEP are reduced, indicating no or a much lower degree of overfitting. Since the reproducibility of the reference method used for dataset A is assumed to be worse than that for dataset B, overfitting during PLS calibration may occur to establish the correlation between the spectral variation and the relatively inaccurate reference concentrations. Even under the same condition, an overfit-free or at least overfit-minimized modeling is possible with the use of RF.

In the case of using naphtha dataset B, there is no significant difference in SEPs between the two methods for the determination of total paraffin concentration. For the measurement of total aromatic concentration, the SEP derived using RF is slightly better than that derived using PLS, while the SEP derived using PLS is slightly better than that derived using RF for the determination of total naphthene concentration. It is difficult to justify the higher error in RF for this case. Nonetheless, RF is a promising method that can improve the accuracy of quantitative analysis for complex samples without overfitting.

4.3. Evaluation of variable importance

To provide further insight into RF modeling, it is valuable to examine the so called variable importance, which indicates the importance of variables at each wavelength (wavenumber) for the construction of RF model [15]. The scheme shown in Fig. 1 is helpful to illustrate the concept of variable importance. For example, suppose that 4200 cm^{-1} is selected at a node in a tree. As noted previously, there are 8 absorbance values from 8 spectra at 4200 cm^{-1}. The order of these 8 absorbance values is initially randomized for the calculation of variable importance at 4200 cm^{-1}. Then, these randomized values are re-fed into a model and the resulting error is calculated. Note that only the order of variables (absorbance values) at 4200 cm^{-1} is randomized while the orders of variables at other wavenumbers remain unchanged. The difference in error before and after the randomization of variable order is the variable importance for a given wavelength. If the difference in error is not significant, then variables at this wavelength do not significantly contribute to the description of concentration variation. On the other hand, variables resulting in large differences in error play important roles in accurate determination of concentrations.

Fig. 5 shows the variable importance in the 4800–4000 cm^{-1} range for the determination of RONs for gasoline samples. Some of representative C_6–C_9 pure hydrocarbons in gasoline are also displayed for comparison. NIR spectra of pure hydrocarbons are arbitrarily offset. Variable importance is high in the 4220–4160 cm^{-1} range and moderate in the

Table 2
PLS and RF results for the determination of total paraffin, total naphthene and aromatic concentrations using datasets A and B.

					Unit: wt%		
		Dataset A			Dataset B		
Method		Number of factors	SEC	SEP	Number of factors	SEC	SEP
PLS	Total paraffin	5	0.76	0.97	5	0.40	0.30
	Total naphthene	5	0.43	0.55	5	0.36	0.26
	Total aromatic	5	0.42	0.50	5	0.25	0.22
Method		Number of trees	SEC	SEP	Number of trees	SEC	SEP
RF	Total paraffin	2000	0.74	0.79	2000	0.56	0.29
	Total naphthene	2000	0.51	0.51	2000	0.52	0.35
	Total aromatic	2000	0.39	0.44	2000	0.21	0.16

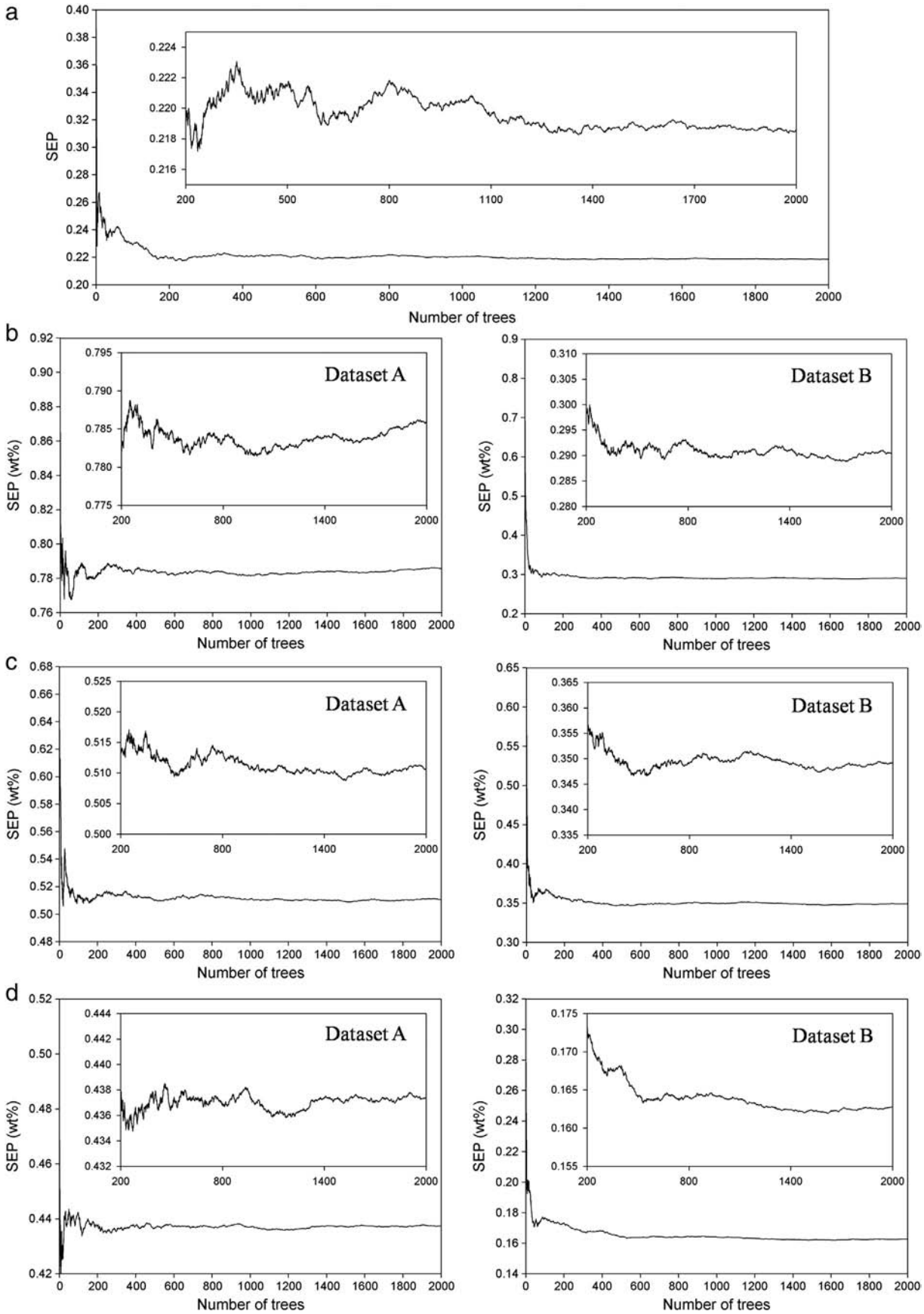

Fig. 4. Variation in SEP as a function of the number of trees (up to 2000 trees) in RF modeling for the determination of RONs (a) for gasoline samples, and total paraffin (b), total naphthene (c) and total aromatic (d) concentrations for naphtha samples. For detail examination, the lower SEP range is enlarged in each figure.

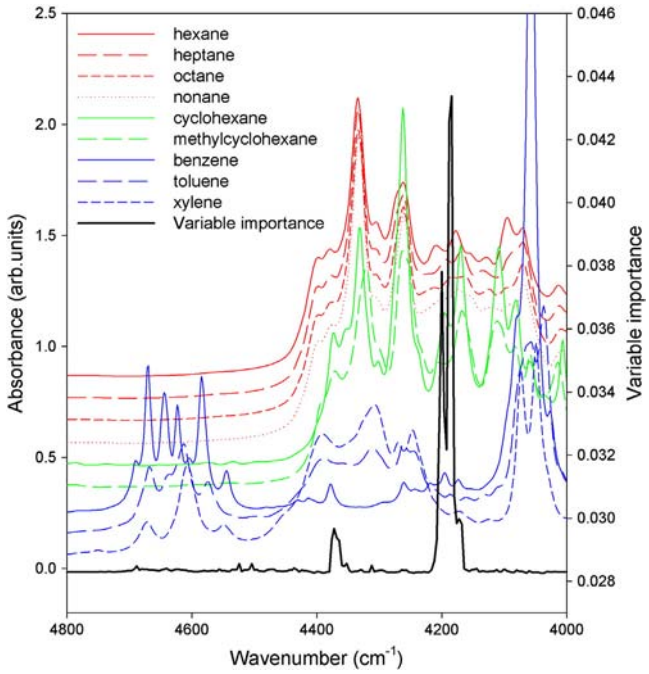

Fig. 5. Variable importance in the 4800–4000 cm^{-1} range for the determination of RONs for gasoline samples. Some of C_6–C_9 pure hydrocarbons are also displayed for comparison.

4380–4350 cm^{-1} range. The spectral ranges showing significant variable importance are fairly localized as shown. It is hard to justify the localized variable importance since spectral features of individual components in gasoline are complexly overlapped each other. However, it clearly indicates that valuable spectral information necessary to describe the variation of RON by each tree is centered at these ranges.

Fig. 6 shows the variable importance in the 4800–4000 cm^{-1} range for the determination of total paraffin (a), total naphthene (b) and total aromatic (c) concentrations using naphtha dataset A. NIR spectra of representative pure components are also displayed for comparison. *n*-Hexane, *n*-heptane, *n*-octane and *n*-nonane are selected for paraffin (a). Cyclohexane and methylcyclohexane (b) are selected for naphthenes, and benzene, toluene and xylene represent aromatic components (c). Variable importance is very high around 4343 cm^{-1} for total paraffin concentration, and peaks of variable importance are also observed at 4609, 4243, 4101, and 4045 cm^{-1} with moderate intensities. The locations of variable importance are close to the band positions of paraffin components, except at 4609 cm^{-1}. Around 4609 cm^{-1}, there are no absorption peaks originating from paraffin or naphthene components, and only NIR bands of aromatic components are present as discussed in Fig. 3. Therefore, the variable importance should be absent around 4609 cm^{-1} for the determination of total paraffin concentration. The observed variable importance around 4609 cm^{-1} may originate from the degraded accuracy of the reference method employed for dataset A. Since uncertainty in the reference concentration is higher, the chance of including misleading variables from non-relevant wavenumbers in a RF model may increase.

In the case of RF modeling for total naphthene concentration, variable importance is substantially high around 4345, 4273 and 4119 cm^{-1}, and these wavenumbers match well with the positions of major absorption bands for cyclohexane and methylcyclohexane. In the case of RF modeling for total aromatic concentration, variable importance around 4601 cm^{-1} is exceptionally high. The bands in the 4700–4500 cm^{-1} range are solely from aromatic components without overlapping with those from other components. Therefore, the high variable importance around 4601 cm^{-1} is reasonable. Wavenumbers around 4101, 4093 and 4031 cm^{-1} show minor variable importance.

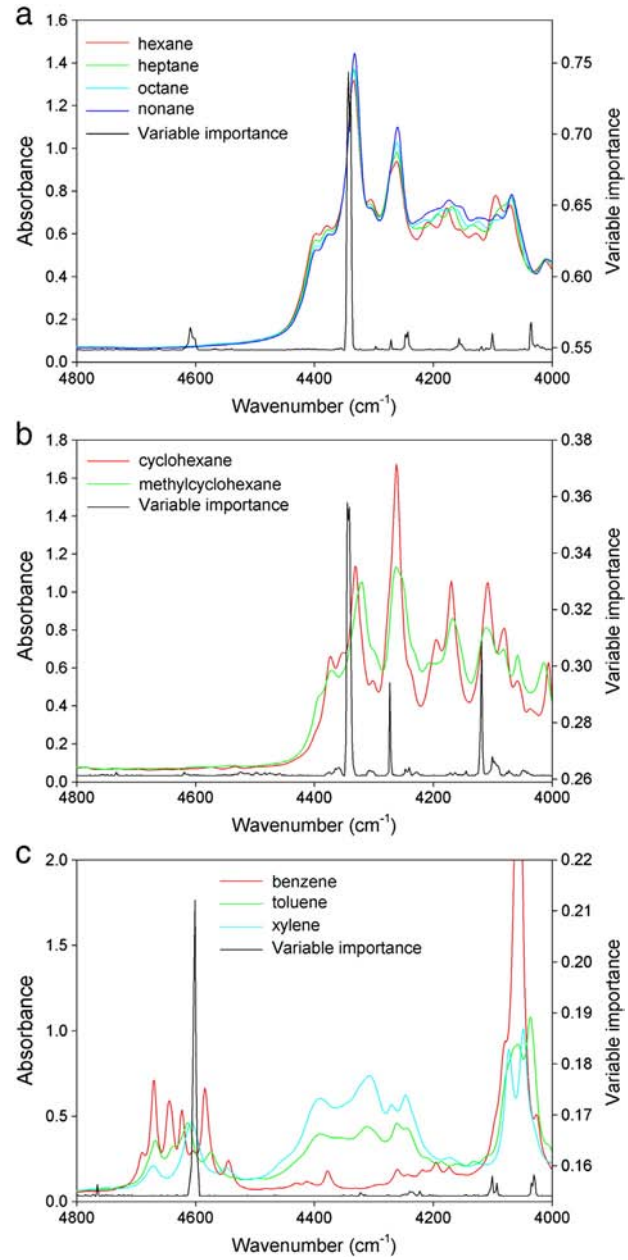

Fig. 6. Variable importance in the 4800–4000 cm^{-1} range for the determination of total paraffin (a), total naphthene (b) and total aromatic (c) concentrations using naphtha dataset A. For comparison, NIR spectra of representative pure components are also displayed.

Fig. 7 shows the variable importance plots resulting from the use of naphtha dataset B. In the case of total paraffin, the variable importance observed around 4609 cm^{-1} completely disappears (Fig. 6(a)). Since the reference method used for dataset B is more accurate than that used for dataset A, the wavenumbers of greater variable importance are within the range of paraffin bands. The patterns of variable importance acquired from naphtha datasets A and B are similar with each other. In each case, the wavelengths corresponding to the most significant variable importance are nearly identical for both datasets, while those of moderate variable importance are slightly different.

4.4. Variation of accuracy depends on the number of training samples

With the use of both PLS and RF, the accuracies were evaluated by varying the number of training samples. Since RF models require more

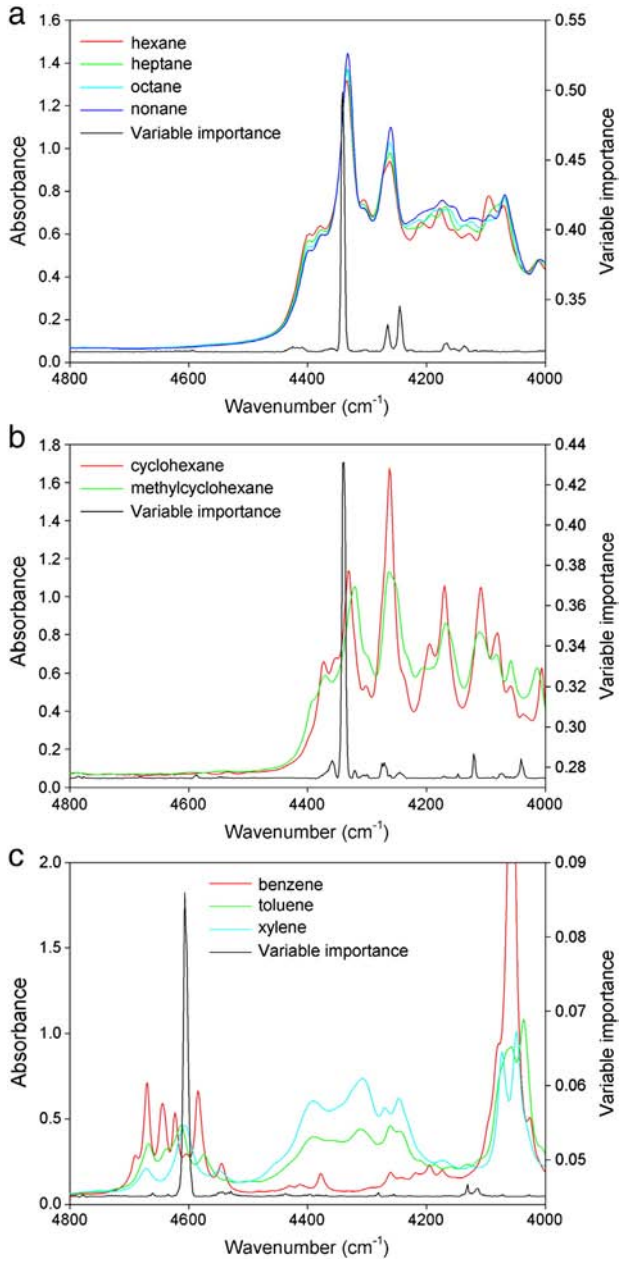

Fig. 7. Variable importance in the 4800–4000 cm^{-1} range for the determination of total paraffin (a), total naphthene (b) and total aromatic (c) concentrations using naphtha dataset B. For comparison, NIR spectra of representative pure components are also displayed.

closely spaced terminal nodes in a tree for accurate analysis, the population of training samples could influence on their accuracy. To explore this issue, SEPs were evaluated by varying the number of calibration samples from 101 to 281 for the gasoline dataset (increment interval: 30). The number of samples in the prediction set was unchanged. Analogously, for the same evaluation, the number of calibration samples was varied from 20 to 160 for naphtha dataset A (increment interval: 20), and from 22 to 238 for naphtha dataset B (increment interval: 27). The number of samples in the prediction sets was also unchanged for both datasets. At each number of calibration samples, both PLS and RF models were developed and corresponding SEPs were obtained.

Fig. 8 shows the variation of SEPs as a function of the number of calibration samples for the determination of RONs (a) for gasoline samples, and total paraffin (b), total naphthene (c) and total aromatic (d) concentrations for naphtha samples. Dashed and solid lines correspond to the results from the use of PLS and RF, respectively. For the determination of RON, the PLS SEP decreases slightly with the increase of samples due to the continual inclusion of compositionally diverse samples for the PLS modeling. Meanwhile, the improvement in accuracy is relatively more substantial in the case of RF, indicating the higher sensitivity of RF modeling accuracy to the number of calibration samples.

In the determination for all three compositions of naphtha samples, the accuracies of PLS prediction are fairly constant regardless of the number of calibration samples. In PLS, as long as calibration samples are prepared to fully represent compositional variation, the prediction of concentrations would be accurate. Therefore, how well calibration samples represent overall compositional variation is obviously much more important than the absolute number of samples. Thus, samples that are populated in a narrow range are not desirable for PLS. Since the calibration samples were randomly selected in the full concentration range, they represent wide compositional variation even though fewer samples were selected. Therefore, no significant improvement in SEP is observed even with the increase of calibration samples.

On the contrary, in the case of RF, the prediction error substantially degrades when fewer calibration samples are used, and gradually improves with the further addition of calibration samples. This trend is the same for all three compositions. Suppose that the concentration of an unknown sample is located between two calibrated concentrations. In RF, the determination of concentration is accomplished by matching an unknown sample to one of the calibration concentrations in a tree, as shown in Figs. 1 and 2. Therefore, a tree cannot predict a concentration value that falls in between two adjacent concentrations. For example, two samples of 14.0 and 15.0 wt.% are included in a calibration dataset for RF. When two unknown samples, whose actual concentrations are 14.4 and 14.7 wt.%, are predicted, a RF model would determine the concentrations of these samples as 14.0 and 15.0 wt.%, respectively. If a sample of 14.5 wt.% is included for RF modeling, the concentrations of the same two samples maybe predicted as 14.5 wt.%, thereby improving the accuracy of concentration determination. Therefore, to increase the accuracy of RF model, the inclusion of samples with narrower concentration intervals in a calibration set is necessary. As such, for all three cases (Fig. 8(b), (c) and (d)), the SEPs improve with the further addition of samples in the calibration set.

To confirm the dependency of RF accuracy on the interval of reference values, average intervals in the reference values at different numbers of calibration samples were calculated for both gasoline and naphtha datasets, and their variation was compared with that of RF accuracy. As also shown in Fig. 8, the dotted lines indicate the average interval of reference values.

In the case of gasoline, the average interval of RON values decreases as the number of calibration samples increases up to 161. Then, it is rather consistent afterward. Since the reference RON values determined by a knock engine has an interval of 0.1, RON interval between two adjacent gasoline samples would not change significantly from this point with the inclusion of additional samples in calibration. By contrast, the SEP also decreases as the number of calibration samples increases up to 161 and continues to decrease with the further addition of calibration samples, while the average interval of RON values remains constant from this point. The continual improvement in SEP is mainly attributed to the inclusion of more gasoline samples with diverse chemical compositions, thereby providing valuable information to describe the variation of RON more accurately.

In the case of naphtha, the average intervals for total paraffin, total naphthene and total aromatic concentrations continuously decrease as the number of calibration samples increases; while, the decrease is more substantial when the number of samples in a dataset is small. Since the concentrations have continuum values in a given range, the narrower concentration interval by randomly employing more naphtha samples is valid. The decrease patterns of average concentration interval as a function of the number of calibration samples closely resemble

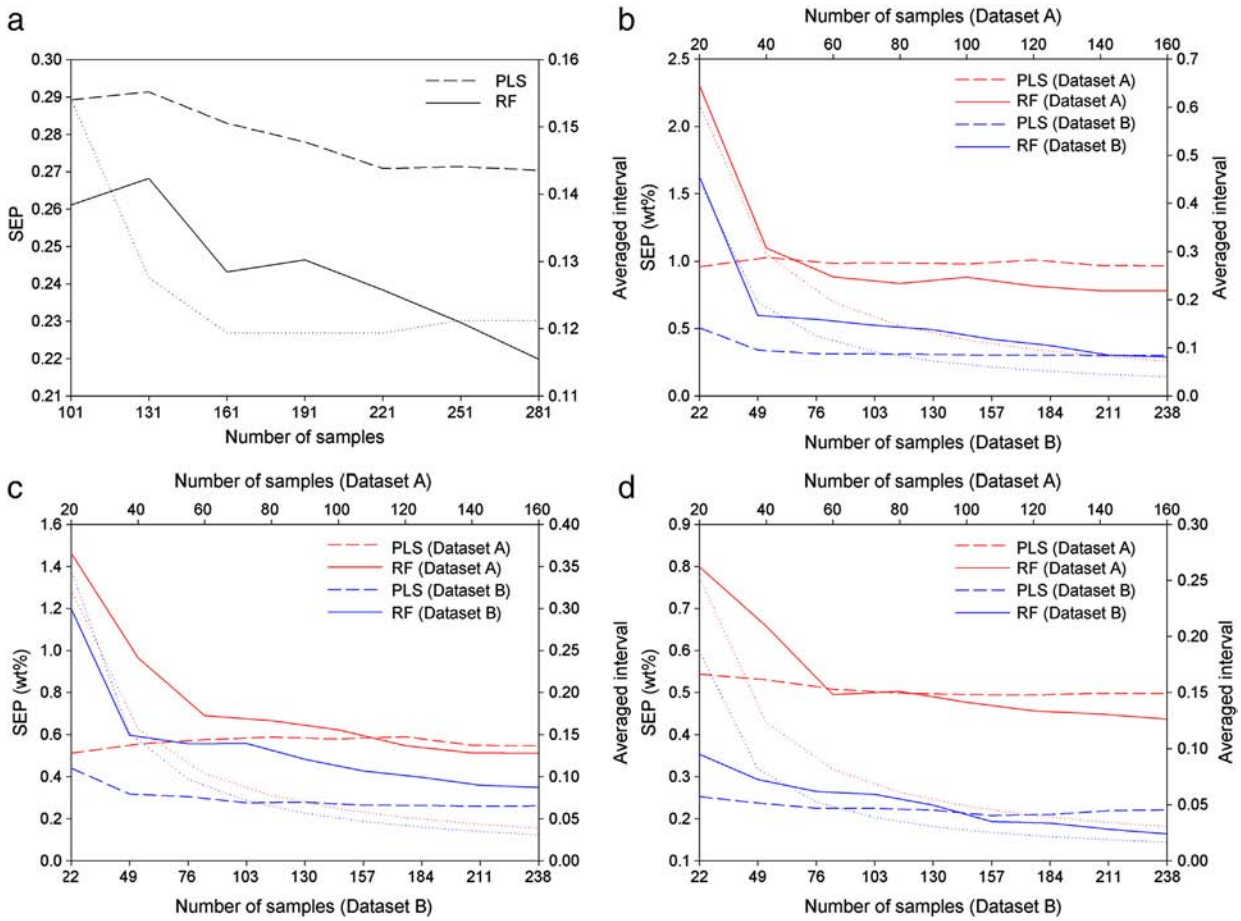

Fig. 8. Variation of SEPs as a function of the number of calibration samples for the determination of RONs (a) for gasoline samples, and total paraffin (b), total naphthene (c) and total aromatic (d) concentrations for naphtha samples. Dashed and solid lines correspond to the result from the use of PLS and RF, respectively. The dotted lines indicate the average interval of reference values.

those of RF SEPs, indicating the importance of preparing samples with narrow concentration intervals to ensure accurate analysis using RF.

5. Conclusions

The potential of RF has been demonstrated as a viable multivariate method for NIR spectroscopic analysis of petroleum-driven products, gasoline and naphtha in this study. Unlike PLS, in the case of RF, calibration samples with narrow concentration intervals across calibration range are necessary for accurate determination of property or concentration. When this requirement is readily achievable, RF could possibly improve accuracy for quantitative analysis of highly complex samples without worry of overfitting, in comparison with PLS. To enhance the utility of RF further, we are currently developing an effective strategy of RF modeling for a situation when the interval for property variation of samples is rather large such as cold properties (cloud and pour point) of diesel.

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